BiTempQA: A Diagnostic Benchmark for Bitemporal Reasoning in LLM Agent Memory Systems

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Abstract

Memory-augmented LLM agents must track not only *what* happened but also *when* it happened and *when they learned about it*, a dual-timestamp (bitemporal) distinction that is important in applications such as medical records, financial systems, and news analysis. Existing benchmarks either ignore timestamps or treat time as a single dimension, leaving this capability insufficiently evaluated.

We introduce BiTempQA, a diagnostic benchmark of 1,536 Chinese question-answer pairs, including 308 held-out test items, spanning 10 scenario types and 9 question types across three difficulty levels. Each memory entry carries explicit event-time and record-time annotations. Over half (56.5%) of questions require reasoning about both timestamps simultaneously. We evaluate five memory backends (FAISS, BM25, Naive RAG, Simple KG, and ChromaDB) in a unified Retrieve $\rightarrow$ LLM Generate $\rightarrow$ LLM Judge pipeline. A mixed-effects logistic regression indicates that record-time reasoning is more difficult than event-time reasoning ($p = 0.023$) after controlling for question type, difficulty, and scenario variation. We further find that (1) the top three systems are not significantly different overall ($p > 0.3$) but diverge by question type (40.6pp accuracy spread); (2) an oracle ensemble reaches 85.4%, a gain of 9.8pp over the best individual system; (3) approximately 55% of sampled failures appear retrieval-related; and (4) system confidence scores are poorly calibrated, with FAISS and BM25 showing AUROC < 0.5. Cross-benchmark comparison with LoCoMo reveals marked system asymmetries: Simple KG performs best on adversarial questions (97.5%) but underperforms on temporal reasoning (37.5%). BiTempQA provides an open diagnostic resource for studying temporally aware memory systems.

1 Introduction

Consider a medical AI assistant that learns at 10:00 AM on Tuesday that a patient's symptoms began two days earlier. To answer "What was the patient's status on Sunday morning?" correctly, the system must distinguish *when the symptoms started* (event time) from *when the system recorded this fact* (record time). This bitemporal distinction, well established in temporal database theory (Snodgrass, 1999; Jensen and Snodgrass, 1999), arises in many settings: financial trades execute on Monday but settle on Wednesday, news events occur on Thursday evening but are reported on Friday morning, and scientific findings may be established in 2022 but published in 2024. A memory system that conflates these timestamps is therefore likely to make incorrect temporal inferences.

Despite its practical importance, this capability is not systematically evaluated in existing benchmarks for LLM-based memory systems. Temporal reasoning benchmarks such as TIME (Wei et al., 2025), ETRQA (Luo et al., 2025), and TEST OF TIME (Palowitch et al., 2025) primarily study reasoning over static text rather than over dynamically evolving memory stores. Memory-system benchmarks such as LoCoMo (Maharana et al., 2024), LongMemEval (Wu et al., 2024), and MemoryAgentBench (Hu et al., 2025) include temporal questions, but they generally treat time as a single dimension and do not distinguish event occurrence from information acquisition. As a result, it remains difficult to determine whether failures on temporal questions arise primarily from *retrieval limitations* or *reasoning limitations*, and which temporal reasoning types are most challenging.

We present BiTempQA, a diagnostic benchmark designed to address this gap. BiTempQA is grounded in a central idea from bitemporal data modeling: each fact has two independent temporal dimensions. The benchmark contains 1,536

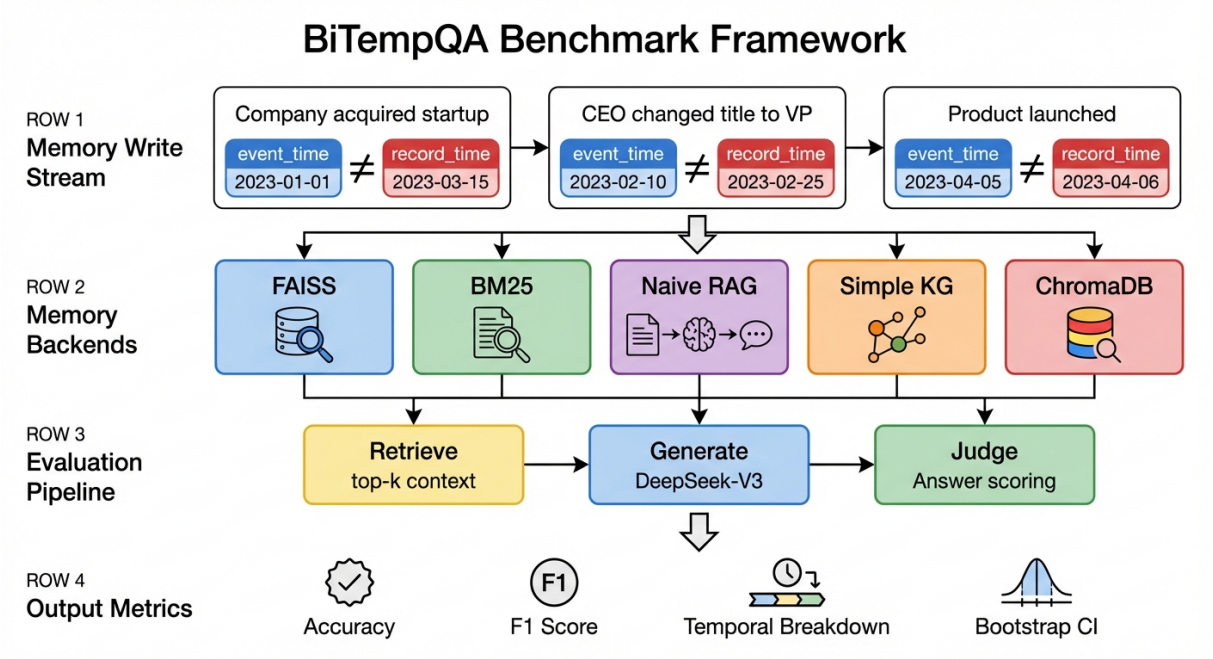

Figure 1: BiTempQA framework overview. Each memory write carries dual timestamps: event time, indicating when the event occurred, and record time, indicating when the system recorded it. Five memory backends retrieve context, which is then processed in a unified Retrieve → Generate → Judge pipeline to produce diagnostic metrics.

Chinese QA pairs, including a 308-item held-out test set, spanning 10 scenario types across 6 domains, with 9 question types at 3 difficulty levels. Each memory write carries explicit event-time and record-time annotations, and 56.5% of questions require reasoning about both timestamps simultaneously. We evaluate five memory backends (FAISS, BM25, Naive RAG, Simple Knowledge Graph, and ChromaDB) in a unified Retrieve → Generate → Judge pipeline.

Our contributions are as follows. First, we introduce, to our knowledge, the first dual-timestamp diagnostic benchmark for memory systems, with explicit event-time and record-time annotations on every memory write and structured metadata on every question. Second, we show that record-time reasoning is associated with lower accuracy than event-time reasoning ($p = 0.023$ in a mixed-effects regression controlling for question type, difficulty, and scenario), supporting the benchmark's diagnostic usefulness. Third, we report several diagnostic findings: sampled failures appear primarily retrieval-related ($\sim 55\%$), confidence is often poorly calibrated (AUROC < 0.5 for some systems), and memory backends exhibit substantial complementarity (oracle +9.8pp over the best single system). Fourth, we compare BiTempQA with LoCoMo (Maharana et al., 2024) and find that graph-based memory is relatively strong on adversarial robustness (97.5%) but comparatively weak on temporal reasoning (37.5%).

2 Related Work

2.1 Temporal Reasoning Benchmarks

Recent years have seen a growing number of temporal reasoning benchmarks for LLMs. TIME (Wei et al., 2025) provides 38,522 questions across 11 subtasks at 3 difficulty levels and evaluates LLMs' ability to understand temporal expressions and reason about event sequences. ETRQA (Luo et al., 2025) consolidates event-temporal reasoning datasets for reading comprehension. TEST OF TIME (Palowitch et al., 2025) introduces synthetic temporal reasoning tasks. MUSTQ (Zhang et al., 2024) evaluates multilingual temporal KGQA across 10 languages. However, all of them treat time as a *single* dimension: none systematically distinguish event occurrence time from information recording time, a distinction that is critical for memory systems.

2.2 Memory System Benchmarks

The emergence of memory-augmented LLM agents has also driven the development of dedicated memory benchmarks. LoCoMo (Maharana et al., 2024) evaluates long conversational memory with 1,986

Benchmark	Year	Temp.	Dual TS	Mem	Lang	Scale
LoCoMo	2024	Partial	✗	✓	EN	1,986
LongMemEval	2024	✗	✗	✓	EN	4,500
TimE	2025	✓	✗	✗	EN	38,522
ETRQA	2025	✓	✗	✗	EN	Consol.
Test of Time	2025	✓	✗	✗	EN	Novel
MusTQ	2024	✓	✗	✗	Multi	15K
BiTempQA	**2026**	✓	✓	✓	ZH	1,536

Table 1: Comparison with related benchmarks. **Dual TS**: evaluates dual-timestamp reasoning. **Mem**: tests memory backends (not just LLMs).

questions across 5 types, including adversarial and temporal categories. LongMemEval (Wu et al., 2024) provides 4,500 questions for evaluating 8 memory systems on single-hop, multi-hop, and temporal reasoning. MemoryAgentBench (Hu et al., 2025) focuses on memory evaluation across multiple datasets. While these benchmarks include temporal questions, they do not systematically test *dual-timestamp* reasoning, namely the ability to distinguish "when this happened" from "when I learned about it."

2.3 Bitemporal Data Modeling

The distinction between event time (valid time) and record time (transaction time) is foundational in temporal database research (Snodgrass, 1999; Jensen and Snodgrass, 1999). Bitemporal data models track both dimensions, enabling queries such as "What did the system know at time T_1 about events that occurred at time T_2?" This concept has been extensively studied in database systems, but it has rarely been brought into the evaluation of LLM memory systems. BiTempQA extends this line of thinking to memory-system evaluation by making bitemporal reasoning an explicit target of assessment.

2.4 Positioning

Table 1 positions BiTempQA relative to existing work. To the best of our knowledge, BiTempQA is currently the only benchmark that both evaluates dual-timestamp reasoning and tests multiple memory-system backends.

3 BiTempQA Benchmark

3.1 Problem Formulation

Following bitemporal data modeling (Snodgrass, 1999), we define two temporal dimensions for memory-system evaluation. *Event time* (t_e) is when an event actually occurred in the real world. *Record time* (t_r) is when the memory system recorded the event. A memory entry is represented as a tuple $(text, t_e, t_r)$, and a memory store at logical time T contains all entries with $t_r \leq T$. A question may require reasoning about t_e only, t_r only, or both simultaneously.

3.2 Scenario Design

We design 10 scenario types spanning 6 domains (corporate, academic, social, fictional, historical, and scientific): (1) Entity Attribute Evolution: attributes change over time (e.g., job titles); (2) Relationship Evolution: social/professional relationships change; (3) Contradictory Information: different sources report conflicting facts; (4) Late-Arriving Facts: events recorded well after they occurred ($|t_r - t_e|$ large); (5) Future-Dated Information: events recorded before their stated occurrence ($t_r < t_e$); (6) Entity Identity Resolution: same entity referenced differently across time; (7) Knowledge Retraction: previously recorded facts are later revoked; (8) Multi-Source Information: multiple independent sources report overlapping events; (9) Gradual Accumulation: knowledge builds incrementally over time; (10) Temporal Ambiguity: unclear or ambiguous temporal references.

Each scenario consists of 3–5 memory writes, each annotated with explicit ISO 8601 timestamps for t_e and t_r. In most cases, scenarios are constructed so that $t_e \neq t_r$, creating natural bitemporal tension.

3.3 Question Taxonomy

Questions are organized into 9 types at 3 difficulty levels:

Level 1 (Easy). Point-in-time retrieval, temporal ordering, first-recorded detection.

Level 2 (Medium). Period queries, change detection, multi-hop temporal reasoning.

Level 3 (Hard). Counterfactual reasoning, version conflict resolution, complex temporal composition.

Questions use three answer formats: multiple choice (4 options, 43.8%), abstractive (free text, 53.0%), and boolean (3.2%). Each question is annotated with metadata indicating whether it requires event time, record time, version tracking, or knowledge retraction.

3.4 Dataset Statistics

BiTempQA contains 1,536 QA pairs across 120 scenarios, split into train (1,075), dev (153), and

Temporal Requirement	Count (%)	Example Question Type
Both timestamps required	174 (56.5%)	"Did the system know X when Y happened?"
Event-time only	110 (35.7%)	"When did event X occur?"
Record-time only	24 (7.8%)	"When was fact X first recorded?"
Version tracking needed	78 (25.3%)	"What is the current value of X?"
Knowledge retraction	28 (9.1%)	"Is the earlier claim about X still valid?"

Table 2: Dual-timestamp composition of the BiTempQA test set (308 questions). Over half of the questions require reasoning about both t_e and t_r. Categories are not mutually exclusive.

test (308). All reported results use the held-out test set (308 pairs across 112 scenarios; 8 scenarios are reserved exclusively for train/dev). Table 2 shows the dual-timestamp composition of the test set.

All QA pairs are generated with Qwen2.5-72B-Instruct using structured prompts and are then validated for parseability and temporal consistency. The dataset is Chinese-language, reflecting the growing importance of multilingual NLP evaluation.

3.5 Evaluation Pipeline

We implement a unified Retrieve → Generate → Judge pipeline (Figure 1):

Step 1: Retrieve. The memory backend retrieves the top-k context passages given the question and the scenario memory store.

Step 2: Generate. DeepSeek-V3 (DeepSeek-AI et al., 2024) generates an answer conditioned on the retrieved context.

Step 3: Judge. An independent LLM call scores the generated answer against the gold answer. For MC questions, exact letter matching is used (100% parse rate). For abstractive and boolean questions, DeepSeek-V3 in "answer_judge" mode evaluates semantic correctness with lenient matching. Without the LLM judge, abstractive accuracy drops to $\sim 10\%$, indicating that exact text matching is too strict; with the judge, accuracy rises to $\sim 75\%$, providing a more semantically appropriate evaluation.

4 Experimental Setup

4.1 Memory Backends

We evaluate five memory backends spanning different retrieval paradigms. FAISS (Douze et al., 2024) uses dense vector similarity search with SiliconFlow's multilingual embedding API and top-$k = 5$ retrieval. BM25 uses sparse keyword matching via rank-BM25 with top-$k = 5$ retrieval. Naive RAG uses sequential passage concatenation (no relevance ranking) with top-$k = 5$. Simple Knowledge Graph is an in-memory graph with entity/relation extraction, returning connected subgraphs (5–10 facts per query). ChromaDB (Chroma, 2023) is a persistent vector database that uses its default embedding function (all-MiniLM-L6-v2).

All systems share identical chunking, top-k parameters, generation prompts, and the same LLM backbone (DeepSeek-V3 via the SiliconFlow API), so observed differences are more plausibly attributable to retrieval design choices.

4.2 Cross-Benchmark Evaluation

We additionally evaluate all backends except ChromaDB on the LoCoMo benchmark (Maharana et al., 2024), which contains 1,986 multiple-choice questions across 10 long conversations (300–700 turns each). This enables a cross-benchmark comparison of system behavior under different conditions: BiTempQA's short, dual-timestamp scenarios versus LoCoMo's long, largely single-timestamp conversations.

4.3 Statistical Methods

We employ four statistical procedures: (1) bootstrap 95% confidence intervals (10,000 iterations); (2) McNemar's test with continuity correction for pairwise system comparisons; (3) mixed-effects logistic regression with scenario random intercepts to control for question type, difficulty, and temporal requirements; and (4) LLM-judge cross-validation (DeepSeek-V3 vs. Qwen2.5-72B, Cohen's $\kappa = 0.455$).

5 Results

5.1 Overall Performance

Table 3 presents overall accuracy by system. FAISS achieves the highest accuracy at 75.6%, followed closely by Naive RAG (74.0%) and BM25 (73.4%). Bootstrap 95% CIs overlap substantially among the top three, and pairwise McNemar's tests show no significant differences ($p > 0.3$). Simple KG (67.2%) and ChromaDB (41.2%) perform significantly worse than FAISS ($p < 0.01$ and $p < 0.001$, respectively). ChromaDB uses a different default embedding model (all-MiniLM-L6-v2) than the other vector-based systems; we include it for completeness but note this confound.

System	Accuracy	95% CI	F1	Ver. Recall
FAISS	**75.6%**	[70.8, 80.2]	**0.870**	**79.5%**
Naive RAG	74.0%	[69.2, 78.9]	0.867	74.4%
BM25	73.4%	[68.2, 78.2]	0.867	69.2%
Simple KG	67.2%	[61.7, 72.4]	0.841	75.6%
ChromaDB†	41.2%	[35.7, 46.8]	0.744	37.2%

Table 3: Overall results on BiTempQA (308 questions). † ChromaDB uses a different embedding model; see §8.

System	Event-Only (110Q)	Record-Only (24Q)	Both Req. (174Q)	Ver. Track. (78Q)
FAISS	79.1%	66.7%	**74.7%**	**79.5%**
Naive RAG	78.2%	62.5%	73.0%	74.4%
BM25	77.3%	66.7%	71.8%	69.2%
Simple KG	66.4%	58.3%	69.0%	75.6%
ChromaDB	44.5%	50.0%	37.9%	37.2%
Average	**69.1%**	60.8%	65.3%	67.2%

Table 4: Accuracy by temporal requirement. Record-only questions ($n = 24$) are hardest across all systems, though the sample is small.

5.2 Record-Time Reasoning Is Genuinely Harder

Table 4 shows accuracy by temporal requirement. Record-only questions average 60.8% compared with 69.1% for event-only questions, yielding an 8.3pp gap that appears consistently across all five systems. To examine whether this pattern can be explained by other dataset factors, we fit a mixed-effects logistic regression with scenario random intercepts, controlling for system identity, question type, difficulty level, and temporal requirement flags.

The record-time coefficient ($\beta = -0.079$, $p = 0.023$) remains significant after controlling for the other factors in the model (Table 5), indicating that record-time requirements are associated with greater difficulty. Notably, difficulty levels (L2/L3) are not independently predictive after controls ($p = 0.103$, $p = 0.278$), suggesting that the observed difficulty gradient is largely mediated by the distribution of harder question types at higher levels.

5.3 Per-Question-Type Analysis

Figure 2 reveals substantial system × type interactions. FAISS leads on complex temporal (87.1%) and counterfactual (53.8%); BM25 excels at point-in-time (85.7%) and multi-hop (83.3%); Simple KG tops period query (88.9%). The 40.6pp spread across question types (counterfactual: 48.7% to temporal order: 89.3%) suggests that the typology captures meaningfully different challenges.

Predictor	Coef.	z	p
System effects (ref: FAISS)			
ChromaDB	−0.344	−10.53	< 0.001
Simple KG	−0.084	−2.58	0.010
BM25	−0.023	−0.70	0.487
Naive RAG	−0.016	−0.50	0.619
Temporal requirements			
Record-time required	−0.079	−2.26	**0.023**
Event-time required	0.061	1.08	0.280
Version tracking	0.050	1.29	0.199
Knowledge retraction	−0.080	−1.50	0.135
Scenario variance	$\sigma^2 = 0.049$		

Table 5: Mixed-effects logistic regression ($N = 1{,}540$) with scenario random intercepts. Record-time requirement is the only significant temporal factor.

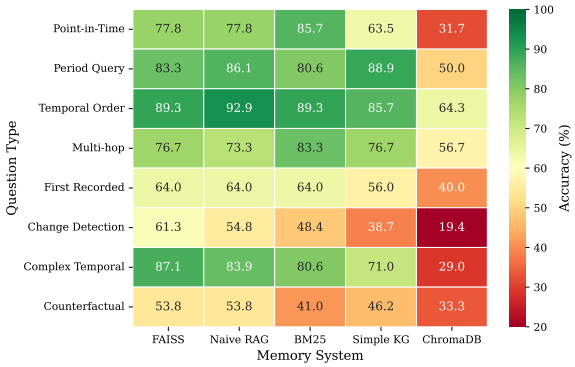

Figure 2: System × question-type accuracy heatmap. No single system dominates all types, and the 40.6pp spread confirms that the typology captures distinct challenges.

5.4 Difficulty Calibration

All five systems show monotonic L1 → L3 degradation, but the magnitude of the drop varies substantially: FAISS declines by only 2.9pp, whereas BM25 declines by 16.6pp (Figure 3). FAISS's dense representations appear to maintain semantic relevance even under complex compositional demands, while BM25's keyword matching deteriorates sharply when query–document lexical overlap decreases.

5.5 System Complementarity

An oracle ensemble (counting an item as correct if any system answers it correctly) reaches 85.4%, a 9.8pp gain over the best single system, suggesting that the memory backends have substantially non-overlapping error profiles (Figure 4). FAISS-ChromaDB Jaccard overlap is only 0.446 (lowest), while FAISS-SimpleKG is 0.746. Simple KG uniquely solves 6 items that FAISS misses, and

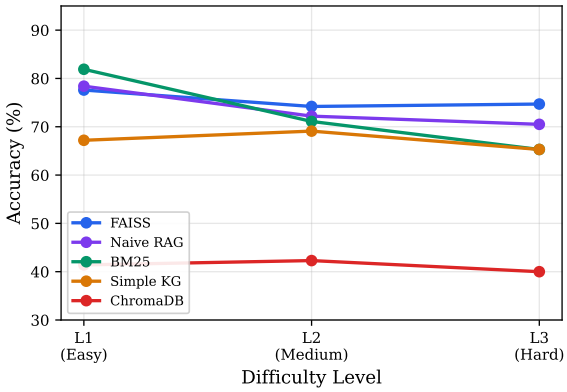

Figure 3: Accuracy degradation from L1 to L3. FAISS is most robust (-2.9pp), whereas BM25 is most fragile (-16.6pp).

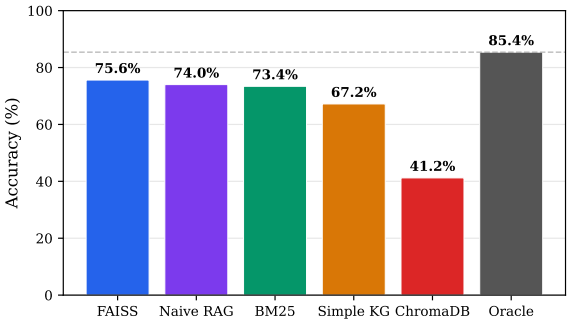

Figure 4: The oracle ensemble reaches 85.4%, versus 75.6% for the best single system. Forty-five irreducible failures suggest retrieval-independent challenges.

ChromaDB uniquely solves 6, despite its low overall accuracy. Forty-five items (14.6%) are answered incorrectly by all systems, indicating a substantial residual error set under the current evaluation setup.

5.6 Cross-Benchmark Validation with LoCoMo

Table 6 compares BiTempQA results with LoCoMo evaluation. The clearest cross-benchmark asymmetry appears in Simple KG: it reaches 97.5% adversarial accuracy on LoCoMo (best among the evaluated systems, +34pp above average), but only 37.5% on LoCoMo temporal reasoning and 67.2% on BiTempQA. This pattern is consistent with a trade-off in which graph structures help avoid adversarial hallucination by returning little or no context for non-existent facts, but offer less retrieval breadth for temporal reasoning.

	BiTempQA		LoCoMo		
System	Overall	Temporal	Overall	Temporal	Adversarial
FAISS	**75.6%**	**75.6%**	**57.0%**	**46.9%**	63.0%
Naive RAG	74.0%	74.0%	54.4%	42.7%	69.7%
BM25	73.4%	73.4%	55.8%	43.8%	64.8%
Simple KG	67.2%	67.2%	56.8%	37.5%	**97.5%**

Table 6: Cross-benchmark comparison. Simple KG shows an extreme adversarial-temporal asymmetry.

System	ECE↓	Brier↓	AUROC
FAISS	0.163	0.229	0.453
Naive RAG	0.319	0.303	0.552
BM25	0.235	0.769	0.467
Simple KG	0.672	0.672	0.500
ChromaDB	0.412	0.412	0.500

Table 7: Confidence calibration metrics. FAISS and BM25 show AUROC < 0.5, meaning that wrong answers receive higher confidence than correct ones.

6 Diagnostic Analysis

6.1 Retrieval Is the Bottleneck

We sampled 20 FAISS failures and grouped them by retrieval-context length. All 20 had short or empty retrieval context ($<$ 200 characters), compared with an average of $\sim$ 500 characters for correct answers. This heuristic analysis attributes roughly $\sim 55\%$ of the sampled failures to retrieval and $\sim 45\%$ to downstream reasoning, suggesting that improvements in retrieval quality, particularly temporal-aware retrieval, may yield larger gains than improvements in reasoning alone.

6.2 Confidence Is Anti-Calibrated

A notable diagnostic finding is that some vector-based systems exhibit anti-calibrated confidence scores. Table 7 shows Expected Calibration Error (ECE), Brier score, and AUROC for each system.

FAISS's AUROC of 0.453 indicates that confidence scores are worse than random at predicting correctness: wrong answers often receive higher confidence than correct ones. This has practical implications, because agents cannot reliably use internal confidence alone to determine whether a temporal-memory answer should be trusted or externally verified.

6.3 Knowledge Retraction vs. Version Tracking

We decompose performance on questions involving knowledge dynamics (Table 8). Simple KG shows an asymmetric profile: it is relatively strong on

System	Retraction (28Q)	Ver. Track. (58Q)	Neither (222Q)
FAISS	67.9%	**82.8%**	74.8%
Naive RAG	67.9%	75.9%	74.3%
BM25	57.1%	74.1%	**75.2%**
Simple KG	53.6%	81.0%	65.3%
ChromaDB	39.3%	37.9%	42.3%

Table 8: Accuracy by knowledge-dynamics type. Simple KG is strong at version tracking but weak at retraction. Bootstrap CIs for retraction span 35pp because $n = 28$ is small.

version tracking (81.0%, close to FAISS's 82.8%) but weaker on knowledge retraction (53.6%). This pattern suggests that graph structures may naturally support tracking fact evolution, while handling explicit revocation of superseded knowledge remains more difficult.

6.4 Implications for Memory System Design

Our findings suggest three design implications:

1. Temporal-aware retrieval. Since $\sim 55\%$ of sampled failures are retrieval-limited, adding temporal metadata filtering (e.g., "retrieve only facts recorded before T") or bitemporal indexing may yield larger gains than improving the reasoning model alone.

2. Hybrid vector+graph architectures. The complementarity between systems (oracle +9.8pp) and Simple KG's adversarial robustness suggest that combining dense retrieval's breadth with graph structure's precision could outperform either approach alone.

3. External verification for temporal claims. Anti-calibrated confidence means that internal signals are unreliable for identifying uncertain temporal answers. External verification, such as cross-checking against source timestamps, may therefore be especially valuable for bitemporal queries.

7 Conclusion

We presented BiTempQA, a diagnostic benchmark for evaluating bitemporal reasoning, that is, reasoning about *when events occurred* versus *when the system learned about them*, in LLM agent memory systems. Our evaluation of five memory backends yields three main findings.

First, our results indicate that record-time reasoning is more difficult than event-time reasoning ($p = 0.023$), supporting the usefulness of the dual-timestamp distinction. Second, retrieval appears to be a major bottleneck ($\sim 55\%$ of sampled failures), suggesting that temporally aware retrieval mechanisms may be a productive direction for improvement. Third, the evaluated memory backends are complementary: an oracle ensemble reaches 85.4% (+9.8pp), and graph-based memory shows strong adversarial robustness (97.5%) despite lower temporal accuracy.

Taken together, these findings suggest several directions for temporally aware memory systems, including bitemporal indexing, hybrid retrieval architectures that combine vector breadth with graph precision, and external verification beyond sole reliance on internal confidence.

Future work includes human expert validation, English translation, evaluation of production memory systems (e.g., Mem0, Graphiti, and TMG), and extension to longer scenarios with richer temporal dynamics.

8 Limitations

We note several limitations of the current benchmark and evaluation setup:

Scale and language. BiTempQA contains 1,536 items (308 test items) and is currently limited to Chinese. While this scale supports diagnostic analysis (bootstrap CIs $\leq$ 8pp for major categories), smaller subcategories (record-only $n = 24$, retraction $n = 28$) have wide confidence intervals that limit strong quantitative claims. Expanding the test set to 600–1000 items and adding English translations are important directions for future work.

LLM-generated data. All QA pairs are generated by Qwen2.5-72B-Instruct and have not yet undergone human expert validation. We mitigate this through (1) structured generation with strict output schemas, (2) automated consistency checks, and (3) LLM Judge cross-validation ($\kappa = 0.455$). However, some items may still contain ambiguous or imperfectly specified answers. Human validation on a stratified sample (100–150 items, 2 native annotators) is therefore a high-priority next step.

ChromaDB confound. ChromaDB uses a different default embedding model (all-MiniLM-L6-v2, 22M parameters) than the other vector systems (SiliconFlow API, multilingual). Its poor performance (41.2%) likely reflects this configuration difference rather than a fundamental limitation of persistent vector databases. We therefore include ChromaDB

for completeness but do not rely on it for the paper's primary comparative claims.

Train/test scenario overlap. All 112 test scenarios also appear in the training set, albeit with different questions. While this mirrors common reading-comprehension setups (same passages, different questions), it also means that systems fine-tuned on the training data may benefit from scenario familiarity. Future versions should use fully held-out scenarios for the test set.

Backend scope. We evaluate memory storage and retrieval backends rather than full agent memory systems (e.g., Mem0, Zep, and Graphiti). These production systems incorporate additional components, such as memory consolidation, summarization, and conflict resolution, that may affect bitemporal reasoning. Evaluating such systems remains ongoing work.

LLM Judge leniency. DeepSeek-V3 is more lenient than Qwen2.5-72B (80% vs. 60% accuracy on a 50-item sample), with 11 of 12 disagreements favoring acceptance. Reported accuracies may therefore be somewhat higher than those obtained with a stricter judge, although relative system rankings are likely to be more stable.

9 Ethics Statement

BiTempQA is intended as an evaluation benchmark for memory systems rather than as a deployed decision-making tool. The dataset is generated from synthetic scenarios and does not intentionally include private personal data. This design reduces direct privacy risk, but it does not eliminate the possibility that generated examples may still reflect cultural, linguistic, or model-induced biases.

Our experiments also rely on LLM-generated QA pairs and an LLM-based judge. As discussed in the limitations section, this introduces risks of annotation noise, judge leniency, and benchmark artifacts. We therefore view the benchmark as a diagnostic instrument for controlled comparison rather than as a definitive measure of real-world temporal reliability.

We recommend that future releases include human validation, broader linguistic coverage, and additional checks for harmful stereotypes or misleading temporal claims. Any use of the benchmark to evaluate production systems in sensitive domains should be accompanied by task-specific safety review and external verification.

References

Chroma. 2023. Chromadb: The ai-native open-source embedding database. Accessed 2026-04-07.

DeepSeek-AI and 1 others. 2024. DeepSeek-V3 technical report. *Preprint*, arXiv:2412.19437.

Matthijs Douze, Alexandr Guzhva, Chengqi Deng, Jeff Johnson, Gergely Szilvasy, Pierre-Emmanuel Mazaré, Maria Lomeli, Lucas Hosseini, and Hervé Jégou. 2024. Faiss library. *Preprint*, arXiv:2401.08281.

Yuanzhe Hu, Yu Wang, and Julian McAuley. 2025. Evaluating memory in LLM agents via incremental multi-turn interactions. *Preprint*, arXiv:2507.05257.

Christian S. Jensen and Richard T. Snodgrass. 1999. Temporal data management. *IEEE Transactions on Knowledge and Data Engineering*, 11(1):36–44.

Sigang Luo, Yinan Liu, Dongying Lin, Yingying Zhai, Bin Wang, Xiaochun Yang, and Junpeng Liu. 2025. ETRQA: A comprehensive benchmark for evaluating event temporal reasoning abilities of large language models. In *Findings of the Association for Computational Linguistics: ACL 2025*, pages 23321–23339, Vienna, Austria. Association for Computational Linguistics.

Adyasha Maharana, Dong-Ho Lee, Sergey Tulyakov, Mohit Bansal, Francesco Barbieri, and Yuwei Fang. 2024. Evaluating very long-term conversational memory of LLM agents. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 13851–13870, Bangkok, Thailand. Association for Computational Linguistics.

John Palowitch, Jonathan Halcrow, Anton Tsitsulin, Bryan Perozzi, Mehran Kazemi, Karishma Malkan, Sungyong Seo, Jinyeong Yim, and Bahare Fatemi. 2025. Test of time: Benchmarking LLMs on temporal reasoning. Google Research publication page.

Richard T. Snodgrass. 1999. TSQL2: A temporal query language. *ACM Transactions on Database Systems*, 24(1):1–58.

Shaohang Wei, Wei Li, Feifan Song, Wen Luo, Tianyi Zhuang, Haochen Tan, Zhijiang Guo, and Houfeng Wang. 2025. TIME: A multi-level benchmark for temporal reasoning of LLMs in real-world scenarios. *Preprint*, arXiv:2505.12891.

Di Wu, Hongwei Wang, Wenhao Yu, Yuwei Zhang, Kai-Wei Chang, and Dong Yu. 2024. LongMemEval: Benchmarking chat assistants on long-term interactive memory. *Preprint*, arXiv:2410.10813.

Tingyi Zhang, Jiaan Wang, Zhixu Li, Jianfeng Qu, An Liu, Zhigang Chen, and Hongping Zhi. 2024. MusTQ: A temporal knowledge graph question answering dataset for multi-step temporal reasoning. In *Findings of the Association for Computational Linguistics: ACL 2024*, pages 11688–11699, Bangkok, Thailand. Association for Computational Linguistics.

A Appendix

A.1 Full Per-Question-Type Results

Table 9 provides complete per-question-type accuracy across all systems and types.

Question Type (Level)	FAISS	BM25	Naive RAG	Simple KG	ChromaDB
Point-in-Time (L1)	77.8%	**85.7%**	77.8%	63.5%	31.7%
Temporal Order (L1)	89.3%	89.3%	**92.9%**	85.7%	64.3%
First Recorded (L1)	64.0%	64.0%	64.0%	56.0%	40.0%
Period Query (L2)	83.3%	80.6%	**86.1%**	88.9%	50.0%
Change Detection (L2)	**61.3%**	48.4%	54.8%	38.7%	19.4%
Multi-hop Temporal (L2)	76.7%	**83.3%**	73.3%	76.7%	56.7%
Counterfactual (L3)	**53.8%**	41.0%	53.8%	46.2%	33.3%
Complex Temporal (L3)	**87.1%**	80.6%	83.9%	71.0%	29.0%
Average	**74.2%**	71.5%	72.2%	65.8%	40.7%

Table 9: Complete per-question-type accuracy. Version Conflict is excluded because of insufficient samples. The spread ranges from 48.7% (Counterfactual) to 92.9% (Temporal Order).

A.2 LLM Judge Cross-Validation Details

We cross-validated DeepSeek-V3 against Qwen2.5-72B-Instruct on 50 abstractive QA pairs from the FAISS system (random sample, seed=42).

Metric	Value
Agreement rate	76.0% (38/50)
Cohen's κ	0.455 (moderate)
DeepSeek-V3 judged correct	80.0% (40/50)
Qwen2.5-72B judged correct	60.0% (30/50)
Disagreement breakdown (12 items)	
DeepSeek lenient	11/12 (91.7%)
Qwen lenient	1/12 (8.3%)
Pattern: partial answer	6/12
Pattern: answer too short	6/12

Table 10: LLM Judge cross-validation. DeepSeek-V3 is systematically more lenient and accepts partial answers that Qwen rejects.